Diabetes Risk Prediction and Classification Analysis Using Machine Learning

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Introduction and Objective

- Diabetes is a health condition in which blood glucose levels become too high.
- Identifying possible risk early can help people seek medical advice and further testing.
- Machine learning can use health measurements to identify patterns in data.

Aim of this project

To build and compare machine learning models that classify a person as diabetic or non-diabetic using the available health measurements.

Dataset at a Glance

- Dataset: Pima Indians Diabetes dataset (Kaggle).
- Total records: **768**.
- Input variables: **8**.
- Target variable: **Outcome**.
- Outcome 0: non-diabetic; Outcome 1: diabetic.

Class distribution

Class	Records
Non-diabetic (0)	500
Diabetic (1)	268

About 65% non-diabetic and 35% diabetic.

The eight inputs include pregnancies, glucose, blood pressure, skin thickness, insulin, BMI, diabetes pedigree function and age.

Exploratory Data Analysis

- The dataset structure, summary values, and distribution of the target variable were analyzed.
- Some health measurements contained zero values that are not realistic measurements for these variables.
- These zeros were treated as missing values for:
  - Glucose: 5 records Blood pressure: 35 records
  - Skin thickness: 227 records Insulin: 374 records
  - BMI: 11 records
- Insulin and skin thickness had the largest amount of missing information.

Main observation

The data needed careful handling before model training, especially because some columns had many missing values.

Data Preparation

1. Replaced the selected zero values with missing-value markers (NaN).
2. Separated the inputs (X) from the target ($y = \text{Outcome}$).
3. Split the data into 80% training data and 20% test data.
4. Used stratification so that both sets retained a similar share of diabetic and non-diabetic records.
5. Used a pipeline to fill missing values with the training-set median and then scale the features.

Why use a pipeline?

It applies the same preparation steps consistently and helps prevent information from the test set entering the training process.

Models and Evaluation

Six classification models were compared:

- Random Forest
- Decision Tree
- Gradient Boosting
- K-Nearest Neighbours (KNN)
- Support Vector Machine (SVM)
- Naive Bayes

How performance was checked

- Five-fold stratified cross-validation on the training data.
- Final evaluation on a separate, untouched test set.

Metrics: Accuracy (overall correctness), Precision (reliability of positive predictions), Recall (diabetic cases correctly found), F1-score (balance of precision and recall), and ROC-AUC (ability to separate the two classes).

Cross-Validation Results

Mean scores from stratified five-fold cross-validation on the training data:

Model	Acc.	Prec.	Recall	F1	AUC
Random Forest	0.764	0.694	0.584	0.633	0.815
Decision Tree	0.735	0.649	0.547	0.589	0.790
Gradient Boosting	0.756	0.672	0.598	0.631	0.821
KNN	0.735	0.643	0.570	0.601	0.798
SVM	**0.780**	**0.742**	0.584	**0.648**	**0.833**
Naive Bayes	0.762	0.686	**0.598**	0.637	0.828

What do these results tell us?

SVM had the highest mean accuracy, precision, F1-score and ROC-AUC. Gradient Boosting and Naive Bayes had the highest mean recall.

Final Test-Set Results

The separate test set contained 154 records, including 54 diabetic and 100 non-diabetic records.

Model	Accuracy	Recall
Random Forest	0.773	0.611
Decision Tree	**0.786**	**0.685**
Gradient Boosting	0.753	0.556
KNN	0.727	0.556
SVM	0.740	0.556
Naive Bayes	0.701	0.630

Confusion matrix: Decision Tree

It correctly identified 37 of the 54 diabetic records and missed 17. It correctly identified 84 of the 100 non-diabetic records.

The test-set results and cross-validation results are different evaluations. The Decision Tree did best on test-set accuracy and recall, while SVM led on most cross-validation averages.

Key Findings and Limitations

- Glucose was the most important feature in the Random Forest feature-importance analysis, followed by BMI, diabetes pedigree function and age.
- No single model identified every diabetic record correctly.
- Recall was moderate, meaning some diabetic cases were missed.

Limitations

- The dataset is relatively small (768 records).
- Insulin and skin-thickness values were missing in many records.
- The dataset represents a specific population, so results may not generalise to everyone.

Conclusion

This project shows us how to use different machine learning models to predict diabetes from existing patient data. SVM performed best overall in our cross-validation results. At the same time, the project showed that missing data can affect the results.

Thank You